

Preprocessing and data cleaning

- **dataset1**: PSS-10, GAD-7, PHQ-9 totals and survey bands
- **dataset2**: DASS-21 (doubled subscale scores) and parsed demographics

```
import pandas as pd
from pathlib import Path

from utils import (
    print_dataset_summary,
    extract_number,
    age_int_to_bracket,
    cgpa_float_to_band,
    semester_int_to_academic_year,
    dass21_stress_score_to_level,
    dass21_anxiety_score_to_level,
    dass21_depression_score_to_level,
)

# assume notebook is run from project root
ROOT = Path.cwd()
RAW_DATA = ROOT / "raw_data"
DATA = ROOT / "data"
```

load raw tables

```
dataset1 = pd.read_csv(RAW_DATA / "MHP Dataset of University Students_processed.csv")
dataset2 = pd.read_csv(
    RAW_DATA / "Dataset for Analyzing University Students Behaviour.csv",
    sep=";",
    encoding="utf-8",
)

for key, (title, frame) in {
    "dataset1": ("MHP Dataset of University Students", dataset1),
    "dataset2": ("Dataset for Analyzing University Students Behaviour", dataset2),
}.items():
    print_dataset_summary(key, title, frame)
```

```
=====
dataset1 – MHP Dataset of University Students
=====
```

Shape: 2,028 rows, 39 columns

Column names:

1. Age
2. Gender
3. University
4. Department
5. Academic_Year
6. Current_CGPA
7. waiver_or_scholarship
8. PSS1
9. PSS2
10. PSS3
11. PSS4
12. PSS5
13. PSS6
14. PSS7
15. PSS8
16. PSS9
17. PSS10
18. Stress Value
19. Stress Label
20. GAD1
21. GAD2
22. GAD3
23. GAD4
24. GAD5
25. GAD6
26. GAD7
27. Anxiety Value
28. Anxiety Label
29. PHQ1
30. PHQ2
31. PHQ3
32. PHQ4
33. PHQ5
34. PHQ6
35. PHQ7
36. PHQ8
37. PHQ9
38. Depression Value
39. Depression Label

Missing values: none

Per-column type-value summary:

Age

```
dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)
```

Gender

```
dtype=str | enum | 3 distinct labels
missing: 0 (0.00%)
```

University

```

        dtype=str | enum | 15 distinct labels
        missing: 0 (0.00%)
Department
        dtype=str | enum | 12 distinct labels
        missing: 0 (0.00%)
Academic_Year
        dtype=str | enum | 5 distinct labels
        missing: 0 (0.00%)
Current_CGPA
        dtype=str | enum | 6 distinct labels
        missing: 0 (0.00%)
waiver_or_scholarship
        dtype=str | enum | 2 distinct labels
        missing: 0 (0.00%)
PSS1
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS2
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS3
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS4
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS5
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS6
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS7
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS8
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS9
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
PSS10
        dtype=int64 | numeric | range [0, 4]
        missing: 0 (0.00%)
Stress Value
        dtype=int64 | numeric | range [0, 40]
        missing: 0 (0.00%)
Stress Label
        dtype=str | enum | 3 distinct labels
        missing: 0 (0.00%)
GAD1
        dtype=int64 | numeric | range [0, 3]
        missing: 0 (0.00%)
GAD2
        dtype=int64 | numeric | range [0, 3]
        missing: 0 (0.00%)

```

GAD3
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

GAD4
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

GAD5
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

GAD6
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

GAD7
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

Anxiety Value
dtype=int64 | numeric | range [0, 21]
missing: 0 (0.00%)

Anxiety Label
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

PHQ1
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ2
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ3
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ4
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ5
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ6
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ7
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ8
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

PHQ9
dtype=int64 | numeric | range [0, 3]
missing: 0 (0.00%)

Depression Value
dtype=int64 | numeric | range [0, 27]
missing: 0 (0.00%)

Depression Label
dtype=str | enum | 6 distinct labels
missing: 0 (0.00%)

=====

dataset2 – Dataset for Analyzing University Students Behaviour

=====

Shape: 351 rows, 53 columns

Column names:

1. Semester(number like: 1st, 2nd...)
2. Age(Years: 18, 19....)
3. Height(inches like:1 feet)
4. Weight:(in kg)
5. Gender
6. Religion
7. Permanent Residence
8. Current Residence
9. Family Type
10. Number of siblings (numbers: 1, 2...)
11. Father's Occupation
12. Mother's Occupation
13. Do you think your family environment is friendly?
14. Academic performance(CGPA like: 3.5.....)
15. Daily average study hour
16. Daily social media uses hours
17. Relationship Status
18. Smoking status
19. Do you perform religious practice regularly?
20. How much satisfied with your academic activities(i,e...Presentations, Assignments, Tutorials..)
21. Do you do any extra curricular activities?
22. If yes what are that...
23. Do you feel that in the aspect of career building your subject has adequate opportunity in our country?
24. Do you think your subject- related job gets enough social value in our country?
25. Have you ever thought about or attempted to kill yourself?
26. How often have you thought about killing yourself in the past year?
27. Have you ever told someone that you were going to commit suicide, or that you might do it?
28. How likely is it that you will attempt suicide someday?
29. How SATISFIED/dissatisfied are you with your current sleep pattern?
30. To what extent do you consider your sleep problem to INTERFERE with your daily functioning (e.g. daytime fatigue, ability to function at work/daily chores, concentration, memory, mood, etc.).
31. How NOTICEABLE to others do you think your sleeping problem is in terms of impairing the quality of your life?
32. How WORRIED/distressed are you about your current sleep problem?
33. I found it hard to wind down
34. I was aware of the dryness of my mouth
35. I couldn't seem to experience any positive feelings at all
36. I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)
37. I found it difficult to work up the initiative to do things
38. I tended to over-react to situations
- 39.

I experienced trembling (e.g. in the hands)
 40.
 I felt that I was using a lot of nervous energy
 41. I was worried about situations in which I might panic and make a fool of myself
 42. I felt that I had nothing to look forward to
 43. I found myself getting agitated
 44. I found it difficult to relax
 45.
 I felt down-hearted and blue
 46. I was intolerant of anything that kept me from getting on with what I was doing
 47. I felt I was close to panic
 48. I was unable to become enthusiastic about anything
 49. I felt I wasn't worth much as a person
 50. I felt that I was rather touchy
 51. I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)
 52. I felt scared without any good reason
 53. I felt that life was meaningless

Missing values (by column):

If yes what are that...: 13 (3.70%)

I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion): 8 (2.28%)

Academic performance(CGPA like: 3.5.....): 5 (1.42%)

Number of siblings (numbers: 1, 2...): 4 (1.14%)

Total cells missing: 30

Per-column type-value summary:

Semester(number like: 1st, 2nd...)

dtype=str | enum | 53 distinct labels

missing: 0 (0.00%)

Age(Years: 18, 19....)

dtype=str | enum | 16 distinct labels

missing: 0 (0.00%)

Height(inches like:1 feet)

dtype=str | enum | 178 distinct labels

missing: 0 (0.00%)

Weight:(in kg)

dtype=str | enum | 117 distinct labels

missing: 0 (0.00%)

Gender

dtype=str | enum | 2 distinct labels

missing: 0 (0.00%)

Religion

dtype=str | enum | 2 distinct labels

missing: 0 (0.00%)

Permanent Residence

dtype=str | enum | 2 distinct labels

missing: 0 (0.00%)

Current Residence

dtype=str | enum | 3 distinct labels

missing: 0 (0.00%)

Family Type

```

dtype=str | enum | 2 distinct labels
missing: 0 (0.00%)
Number of siblings (numbers: 1, 2...)
dtype=str | enum | 30 distinct labels
missing: 4 (1.14%)
Father's Occupation
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
Mother's Occupation
dtype=str | enum | 3 distinct labels
missing: 0 (0.00%)
Do you think your family environment is friendly?
dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)
Academic performance(CGPA like: 3.5.....)
dtype=str | enum | 144 distinct labels
missing: 5 (1.42%)
Daily average study hour
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
Daily social media uses hours
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
Relationship Status
dtype=str | enum | 3 distinct labels
missing: 0 (0.00%)
Smoking status
dtype=str | enum | 2 distinct labels
missing: 0 (0.00%)
Do you perform religious practice regularly?
dtype=str | enum | 3 distinct labels
missing: 0 (0.00%)
How much satisfied with your academic activities(i,e...Presentations, Assignments, Tu
torials..)
dtype=str | enum | 2 distinct labels
missing: 0 (0.00%)
Do you do any extra curricular activities?
dtype=str | enum | 2 distinct labels
missing: 0 (0.00%)
If yes what are that...
dtype=str | enum | 208 distinct labels
missing: 13 (3.70%)
Do you feel that in the aspect of career building your subject has adequate opportuni
ty in our country?
dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)
Do your think your subject- related job gets enough social value in our country
dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)
Have you ever thought about or attempted to kill yourself?
dtype=str | enum | 6 distinct labels
missing: 0 (0.00%)
How often have you thought about killing yourself in the past year?
dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)
Have you ever told someone that you were going to commit suicide, or that you might d

```

o it?

dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)

How likely is it that you will attempt suicide someday?

dtype=str | enum | 6 distinct labels
missing: 0 (0.00%)

How SATISFIED/dissatisfied are you with your current sleep pattern?

dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)

To what extent do you consider your sleep problem to INTERFERE with your daily functioning (e.g. daytime fatigue, ability to function at work/daily chores, concentration, memory, mood, etc.).

dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)

How NOTICEABLE to others do you think your sleeping problem is in terms of impairing the quality of your life?

dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)

How WORRIED/distressed are you about your current sleep problem?

dtype=str | enum | 5 distinct labels
missing: 0 (0.00%)

I found it hard to wind down

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I was aware of the dryness of my mouth

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I couldn't seem to experience any positive feelings at all

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)

dtype=str | enum | 4 distinct labels
missing: 8 (2.28%)

I found it difficult to work up the initiative to do things

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I tended to over-react to situations

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I experienced trembling (e.g. in the hands)

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I felt that I was using a lot of nervous energy

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I was worried about situations in which I might panic and make a fool of myself

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I felt that I had nothing to look forward to


```

dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I found myself getting agitated
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I found it difficult to relax
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

I felt down-hearted and blue
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I was intolerant of anything that kept me from getting on with what I was doing
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I felt I was close to panic
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I was unable to become enthusiastic about anything
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I felt I wasn't worth much as a person
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I felt that I was rather touchy
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I felt scared without any good reason
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)
I felt that life was meaningless
dtype=str | enum | 4 distinct labels
missing: 0 (0.00%)

```

constants shared by both datasets

```

AGE_ORDER = ["<18", "18-22", "23-26", "27-30", ">30"]
GENDER_LABELS = ["female", "male", "other"]

# department codes for dataset1
# dataset2 uses only "not_recorded" but categories must match
DEPARTMENT_CODES = sorted(
    {
        "bio_sciences",
        "business",
        "eng_cs",
        "eng_civil",
        "eng_eee",
        "eng_mech",
        "eng_other",
        "env_life_sciences",
    }
)

```

```

        "law",
        "liberal_arts",
        "other",
        "pharmacy",
        "not_recorded",
    }
)

ACADEMIC_YEAR_ORDER = ["unknown", "first", "second", "third", "fourth", "other"]
CGPA_ORDER = ["other", "<2.50", "2.50-2.99", "3.00-3.39", "3.40-3.79", "3.80-4.00"]

SEVERITY_ORDER = ["none", "minimal", "mild", "moderate", "moderately_severe", "severe"]

GENDER_FROM_RAW = {
    "Female": "female",
    "Male": "male",
    "Prefer not to say": "other",
}

OUTPUT_COLUMNS = [
    "age",
    "gender",
    "department",
    "academic_year",
    "cgpa",
    "stress_level",
    "stress_z_score",
    "anxiety_level",
    "anxiety_z_score",
    "depression_level",
    "depression_z_score",
]

```

dataset1: column renames, value maps, output schema

```

DATASET1_COLUMN_MAP = {
    "Age": "age",
    "Gender": "gender",
    "Department": "department",
    "Academic_Year": "academic_year",
    "Current_CGPA": "cgpa",
    "Stress Value": "pss10_stress_score",
    "Stress Label": "stress_level",
    "Anxiety Value": "gad7_anxiety_score",
    "Anxiety Label": "anxiety_level",
    "Depression Value": "phq9_depression_score",
    "Depression Label": "depression_level",
}

DATASET1_VALUE_MAPS = {
    "age": {
        "Below 18": "<18",
        "18-22": "18-22",
        "23-26": "23-26",
        "27-30": "27-30",
    }
}

```

```

    "Above 30": ">30",
  },
  "department": {
    "Biological Sciences": "bio_sciences",
    "Business and Entrepreneurship Studies": "business",
    "Engineering – CS / CSE / CSC / Similar to CS": "eng_cs",
    "Engineering – Civil Engineering / Similar to CE": "eng_civil",
    "Engineering – EEE/ ECE / Similar to EEE": "eng_eee",
    "Engineering – Mechanical Engineering / Similar to ME": "eng_mech",
    "Engineering – Other": "eng_other",
    "Environmental and Life Sciences": "env_life_sciences",
    "Law and Human Rights": "law",
    "Liberal Arts and Social Sciences": "liberal_arts",
    "Other": "other",
    "Pharmacy and Public Health": "pharmacy",
  },
  "academic_year": {
    "First Year or Equivalent": "first",
    "Second Year or Equivalent": "second",
    "Third Year or Equivalent": "third",
    "Fourth Year or Equivalent": "fourth",
    "Other": "other",
  },
  "cgpa": {
    "Other": "other",
    "Below 2.50": "<2.50",
    "2.50 – 2.99": "2.50–2.99",
    "3.00 – 3.39": "3.00–3.39",
    "3.40 – 3.79": "3.40–3.79",
    "3.80 – 4.00": "3.80–4.00",
  },
  "stress_level": {
    "Low Stress": "minimal",
    "Moderate Stress": "moderate",
    "High Perceived Stress": "severe",
  },
  "anxiety_level": {
    "Minimal Anxiety": "minimal",
    "Mild Anxiety": "mild",
    "Moderate Anxiety": "moderate",
    "Severe Anxiety": "severe",
  },
  "depression_level": {
    "No Depression": "none",
    "Minimal Depression": "minimal",
    "Mild Depression": "mild",
    "Moderate Depression": "moderate",
    "Moderately Severe Depression": "moderately_severe",
    "Severe Depression": "severe",
  },
}

```

```

DATASET1_OUTPUT_COLUMNS = [
  "age",
  "gender",
  "department",

```

```

    "academic_year",
    "cgpa",
    "pss10_stress_score",
    "stress_level",
    "stress_z_score",
    "gad7_anxiety_score",
    "anxiety_level",
    "anxiety_z_score",
    "phq9_depression_score",
    "depression_level",
    "depression_z_score",
]

```

dataset1: build dataset1_clean

```

dataset1_clean = dataset1[list(DATASET1_COLUMN_MAP.keys())].copy()
dataset1_clean = dataset1_clean.rename(columns=DATASET1_COLUMN_MAP)

dataset1_clean["gender"] = dataset1_clean["gender"].map(GENDER_FROM_RAW)

for col, mapping in DATASET1_VALUE_MAPS.items():
    dataset1_clean[col] = dataset1_clean[col].map(mapping)

for col, cats in (
    ("age", AGE_ORDER),
    ("academic_year", ACADEMIC_YEAR_ORDER),
    ("cgpa", CGPA_ORDER),
    ("stress_level", SEVERITY_ORDER),
    ("anxiety_level", SEVERITY_ORDER),
    ("depression_level", SEVERITY_ORDER),
):
    dataset1_clean[col] = pd.Categorical(
        dataset1_clean[col], categories=cats, ordered=True
    )

dataset1_clean["gender"] = pd.Categorical(
    dataset1_clean["gender"], GENDER_LABELS, ordered=False
)
dataset1_clean["department"] = pd.Categorical(
    dataset1_clean["department"], DEPARTMENT_CODES, ordered=False
)

# calculate z-scores
dataset1_clean["stress_z_score"] = (
    dataset1_clean["pss10_stress_score"] - dataset1_clean["pss10_stress_score"].mean()
) / dataset1_clean["pss10_stress_score"].std()
dataset1_clean["anxiety_z_score"] = (
    dataset1_clean["gad7_anxiety_score"] - dataset1_clean["gad7_anxiety_score"].mean()
) / dataset1_clean["gad7_anxiety_score"].std()
dataset1_clean["depression_z_score"] = (
    dataset1_clean["phq9_depression_score"]
    - dataset1_clean["phq9_depression_score"].mean()
) / dataset1_clean["phq9_depression_score"].std()

```

```
# reorder columns
dataset1_clean = dataset1_clean[DATASET1_OUTPUT_COLUMNS]

dataset1_clean.info()
dataset1_clean.head()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2028 entries, 0 to 2027
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                   2028 non-null   category
1   gender                               2028 non-null   category
2   department                           2028 non-null   category
3   academic_year                       2028 non-null   category
4   cgpa                                 2028 non-null   category
5   pss10_stress_score                  2028 non-null   int64
6   stress_level                        2028 non-null   category
7   stress_z_score                      2028 non-null   float64
8   gad7_anxiety_score                 2028 non-null   int64
9   anxiety_level                      2028 non-null   category
10  anxiety_z_score                    2028 non-null   float64
11  phq9_depression_score              2028 non-null   int64
12  depression_level                   2028 non-null   category
13  depression_z_score                 2028 non-null   float64
dtypes: category(8), float64(3), int64(3)
memory usage: 113.5 KB
```

	age	gender	department	academic_year	cgpa	pss10_stress_score	stress_level	stress_z_s
0	18-22	female	eng_cs	second	2.50-2.99	29	severe	0.88
1	18-22	male	eng_cs	third	3.00-3.39	24	moderate	0.14
2	18-22	male	eng_cs	third	3.00-3.39	15	moderate	-1.18
3	18-22	male	eng_cs	third	3.00-3.39	17	moderate	-0.88
4	18-22	male	eng_cs	second	2.50-2.99	32	severe	1.33

DASS-21 (Depression, Anxiety, and Stress Scale)

The [DASS-21](#) measures Depression, Anxiety, and Stress through 21 items (7 questions per category). Answers are scored 0 to 3.

The rating scale is as follows:

- **0:** Did not apply to me at all — *NEVER*
- **1:** Applied to me to some degree, or some of the time — *SOMETIMES*
- **2:** Applied to me to a considerable degree, or a good part of time — *OFTEN*

- **3:** Applied to me very much, or most of the time — *ALMOST ALWAYS*

Depression (7 items): Measures lack of interest, hopelessness, and dysphoria.

- I couldn't seem to experience any positive feelings at all
- I found it difficult to work up the initiative to do things
- I felt that I had nothing to look forward to
- I felt down-hearted and blue
- I was unable to become enthusiastic about anything
- I felt I wasn't worth much as a person
- I felt that life was meaningless

Anxiety (7 items): Measures physical arousal and situational panic.

- I was aware of the dryness of my mouth
- I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)
- I experienced trembling (e.g. in the hands)
- I was worried about situations in which I might panic and make a fool of myself
- I felt I was close to panic
- I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)
- I felt scared without any good reason

Stress (7 items): Measures tension, agitation, and difficulty relaxing.

- I found it hard to wind down
- I tended to over-react to situations
- I felt that I was using a lot of nervous energy
- I found myself getting agitated
- I found it difficult to relax
- I was intolerant of anything that kept me from getting on with what I was doing
- I felt that I was rather touchy

Severity Level	Depression Score	Anxiety Score	Stress Score
Normal	0 - 9	0 - 7	0 - 14
Mild	10 - 13	8 - 9	15 - 18
Moderate	14 - 20	10 - 14	19 - 25
Severe	21 - 27	15 - 19	26 - 33
Extremely Severe	28+	20+	34+

dataset2: column renames, constants, value maps, output schema

```

# source .csv column names
# DASS items = last 21 columns
DATASET2_COLUMN_MAP = {
    "Semester( number like: 1st, 2nd...)": "semester_raw",
    "Age( Years: 18, 19....)": "age_raw",
    "Gender": "gender",
    "Academic performance(CGPA like: 3.5.....)": "cgpa_raw",
}

# DASS-21 answers to 0-3 (must match .csv wording)
DATASET2_DASS_ANSWER_MAP = {
    "Did not apply to me at all": 0,
    "Applied to me to some degree, or some of the time": 1,
    "Applied to me to a considerable degree or a good part of time": 2,
    "Applied to me very much or most of the time": 3,
}

# standard DASS-21 ordering
DATASET2_DASS_STRESS_INDICES = [1, 6, 8, 11, 12, 14, 18]
DATASET2_DASS_ANXIETY_INDICES = [2, 4, 7, 9, 15, 19, 20]
DATASET2_DASS_DEPRESSION_INDICES = [3, 5, 10, 13, 16, 17, 21]

DATASET2_OUTPUT_COLUMNS = [
    "age_parsed",
    "age",
    "gender",
    "department",
    "semester_parsed",
    "academic_year",
    "cgpa_parsed",
    "cgpa",
    "dass21_stress_score",
    "stress_level",
    "stress_z_score",
    "dass21_anxiety_score",
    "anxiety_level",
    "anxiety_z_score",
    "dass21_depression_score",
    "depression_level",
    "depression_z_score",
]

```

dataset2: build dataset2_intermediate and dataset2_clean

```

dass21_src_cols = list(dataset2.columns[-21:])
DASS_COLS_MAP = {raw: f"dass_{i:02d}" for i, raw in enumerate(dass21_src_cols, start=1)}
dataset2_wide = dataset2[list(DATASET2_COLUMN_MAP.keys()) + dass21_src_cols].copy()
dataset2_wide = dataset2_wide.rename(columns={**DATASET2_COLUMN_MAP, **DASS_COLS_MAP})

print("DASS mapping:")
for k, v in DASS_COLS_MAP.items():
    print(f"- {v}: {k.strip()}")

```

DASS mapping:

- dass_01: I found it hard to wind down
- dass_02: I was aware of the dryness of my mouth
- dass_03: I couldn't seem to experience any positive feelings at all
- dass_04: I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)
- dass_05: I found it difficult to work up the initiative to do things
- dass_06: I tended to over-react to situations
- dass_07: I experienced trembling (e.g. in the hands)
- dass_08: I felt that I was using a lot of nervous energy
- dass_09: I was worried about situations in which I might panic and make a fool of myself
- dass_10: I felt that I had nothing to look forward to
- dass_11: I found myself getting agitated
- dass_12: I found it difficult to relax
- dass_13: I felt down-hearted and blue
- dass_14: I was intolerant of anything that kept me from getting on with what I was doing
- dass_15: I felt I was close to panic
- dass_16: I was unable to become enthusiastic about anything
- dass_17: I felt I wasn't worth much as a person
- dass_18: I felt that I was rather touchy
- dass_19: I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)
- dass_20: I felt scared without any good reason
- dass_21: I felt that life was meaningless

parse age

```
dataset2_wide["age_parsed"] = (  
    dataset2_wide["age_raw"].apply(lambda x: round(extract_number(x))).astype("Int64"  
)  
  
print("mapping: raw age -> parsed age")  
mapping_df = (  
    dataset2_wide[["age_raw", "age_parsed"]].drop_duplicates().sort_values("age_parsed"  
)  
display(mapping_df)
```

mapping: raw age -> parsed age

	age_raw	age_parsed
81	18	18
27	19	19
335	19 years 11 months 24 days	19
22	20	20
334	20 years 9 months 30 days	20
1	21	21
2	22	22
6	22,5	22
4	23	23
0	24	24
213	24+	24
66	25	25
130	Years: 25	25
173	26	26
264	28	28
150	29	29

parse semester

```
# manually defined mapping after inspection of semester values
_WORD_TO_SEMESTER = {
    "second": 2,
    "last": 9, # map to "other"
}

def parse_semester(value: str) -> int | None:
    n = extract_number(value)
    if n is not None:
        return int(n)
    clean = str(value).strip().lower()
    for word, num in _WORD_TO_SEMESTER.items():
        if word in clean:
            return num
    return None

dataset2_wide["semester_parsed"] = (
    dataset2_wide["semester_raw"].apply(parse_semester).astype("Int64")
)

print("mapping: raw semester -> parsed semester")
```

```
mapping_df = (  
  dataset2_wide[["semester_raw", "semester_parsed"]]  
  .drop_duplicates()  
  .sort_values("semester_parsed")  
)  
display(mapping_df)
```

mapping: raw semester -> parsed semester

	semester_raw	semester_parsed
87	1 st	1
65	1st	1
54	2nd	2
199	2nd	2
202	2	2
119	second	2
106	2nd semester	2
317	3rd	3
5	4th	4
251	Complete 4th semester	4
13	4	4
21	4 th	4
313	4th semester	4
57	4semrster	4
0	5th	5
45	5th	5
24	5	5
325	5th Semester	5
340	5'2"	5
183	5th semester	5
52	6 th	6
71	6	6
85	6th semester	6
7	6th	6
47	7	7
6	7 th	7
3	7th	7
344	8th as per old ID. 4th as per new ID. I have c...	8
1	8th	8
123	8th semester	8
18	8	8
333	Last	9

	semester_raw	semester_parsed
220	last	9
132	9	9
305	9th	9
121	9th	9
147	10 th	10
156	10th semester	10
158	10th	10
122	10	10
127	11th	11
169	11	11
140	12	12
179	12th	12
229	13th	13
101	13	13
288	14	14
265	14 th	14
227	14th	14
307	1720 batch	1720
200	Spring-2024	2024
129	Spring 2024	2024
172	summer	<NA>

```
# manual fixes (invalid or year-like semester strings)
BAD_SEM_IDX = [129, 200, 307, 340]
dataset2_wide.loc[BAD_SEM_IDX, "semester_parsed"] = pd.NA
```

parse cgpa

```
dataset2_wide["cgpa_parsed"] = (
    dataset2_wide["cgpa_raw"].apply(extract_number).astype("Float64")
)

print("mapping: raw cgpa -> parsed cgpa")
mapping_df = (
    dataset2_wide[["cgpa_raw", "cgpa_parsed"]]
    .drop_duplicates()
    .sort_values("cgpa_parsed")
```

```
)
display(mapping_df)
```

mapping: raw cgpa -> parsed cgpa

	cgpa_raw	cgpa_parsed
155	00	0.0
205	1,27	1.27
147	1,52	1.52
63	2,09	2.09
104	2,26	2.26
...	...	...
70	NaN	<NA>
144	I am a fresher, first semester student.	<NA>
146	not get yet	<NA>
148	no cgpa	<NA>
290	Yes	<NA>

145 rows x 2 columns

parse DASS scores

```
dass_cols = [c for c in dataset2_wide.columns if c.startswith("dass_")]
for c in dass_cols:
    dataset2_wide[c] = dataset2_wide[c].map(DATASET2_DASS_ANSWER_MAP).astype("Int64")

stress_cols = [f"dass_{i:02d}" for i in DATASET2_DASS_STRESS_INDICES]
anxiety_cols = [f"dass_{i:02d}" for i in DATASET2_DASS_ANXIETY_INDICES]
depression_cols = [f"dass_{i:02d}" for i in DATASET2_DASS_DEPRESSION_INDICES]

dataset2_wide["dass21_stress_score"] = (
    dataset2_wide[stress_cols].sum(axis=1, min_count=7) * 2
)
dataset2_wide["dass21_anxiety_score"] = (
    dataset2_wide[anxiety_cols].sum(axis=1, min_count=7) * 2
)
dataset2_wide["dass21_depression_score"] = (
    dataset2_wide[depression_cols].sum(axis=1, min_count=7) * 2
)

# calculate z-scores
dataset2_wide["stress_z_score"] = (
    dataset2_wide["dass21_stress_score"] - dataset2_wide["dass21_stress_score"].mean()
) / dataset2_wide["dass21_stress_score"].std()
dataset2_wide["anxiety_z_score"] = (
    dataset2_wide["dass21_anxiety_score"] - dataset2_wide["dass21_anxiety_score"].mean()
) / dataset2_wide["dass21_anxiety_score"].std()
```

```
dataset2_wide["depression_z_score"] = (
    dataset2_wide["dass21_depression_score"]
    - dataset2_wide["dass21_depression_score"].mean()
) / dataset2_wide["dass21_depression_score"].std()
```

map parsed values to brackets to match dataset1

```
dataset2_wide["department"] = "not_recorded"
dataset2_wide["gender"] = dataset2_wide["gender"].map(GENDER_FROM_RAW).fillna("other")

dataset2_wide["age"] = dataset2_wide["age_parsed"].map(age_int_to_bracket)
dataset2_wide["cgpa"] = dataset2_wide["cgpa_parsed"].map(cgpa_float_to_band)
dataset2_wide["academic_year"] = dataset2_wide["semester_parsed"].map(
    semester_int_to_academic_year
)

dataset2_wide["stress_level"] = dataset2_wide["dass21_stress_score"].map(
    dass21_stress_score_to_level
)
dataset2_wide["anxiety_level"] = dataset2_wide["dass21_anxiety_score"].map(
    dass21_anxiety_score_to_level
)
dataset2_wide["depression_level"] = dataset2_wide["dass21_depression_score"].map(
    dass21_depression_score_to_level
)

for col, cats in (
    ("age", AGE_ORDER),
    ("academic_year", ACADEMIC_YEAR_ORDER),
    ("cgpa", CGPA_ORDER),
    ("stress_level", SEVERITY_ORDER),
    ("anxiety_level", SEVERITY_ORDER),
    ("depression_level", SEVERITY_ORDER),
):
    dataset2_wide[col] = pd.Categorical(
        dataset2_wide[col], categories=cats, ordered=True
    )

dataset2_wide["gender"] = pd.Categorical(
    dataset2_wide["gender"], categories=GENDER_LABELS, ordered=False
)
dataset2_wide["department"] = pd.Categorical(
    dataset2_wide["department"], categories=DEPARTMENT_CODES, ordered=False
)

dataset2_wide.info()

dataset2_clean = dataset2_wide[DATASET2_OUTPUT_COLUMNS].copy()
dataset2_clean.info()
```

<class 'pandas.DataFrame'>

RangeIndex: 351 entries, 0 to 350

Data columns (total 41 columns):

#	Column	Non-Null Count	Dtype
0	semester_raw	351 non-null	str
1	age_raw	351 non-null	str
2	gender	351 non-null	category
3	cgpa_raw	346 non-null	str
4	dass_01	351 non-null	Int64
5	dass_02	351 non-null	Int64
6	dass_03	351 non-null	Int64
7	dass_04	343 non-null	Int64
8	dass_05	351 non-null	Int64
9	dass_06	351 non-null	Int64
10	dass_07	351 non-null	Int64
11	dass_08	351 non-null	Int64
12	dass_09	351 non-null	Int64
13	dass_10	351 non-null	Int64
14	dass_11	351 non-null	Int64
15	dass_12	351 non-null	Int64
16	dass_13	351 non-null	Int64
17	dass_14	351 non-null	Int64
18	dass_15	351 non-null	Int64
19	dass_16	351 non-null	Int64
20	dass_17	351 non-null	Int64
21	dass_18	351 non-null	Int64
22	dass_19	351 non-null	Int64
23	dass_20	351 non-null	Int64
24	dass_21	351 non-null	Int64
25	age_parsed	351 non-null	Int64
26	semester_parsed	346 non-null	Int64
27	cgpa_parsed	342 non-null	Float64
28	dass21_stress_score	351 non-null	Int64
29	dass21_anxiety_score	343 non-null	Int64
30	dass21_depression_score	351 non-null	Int64
31	stress_z_score	351 non-null	Float64
32	anxiety_z_score	343 non-null	Float64
33	depression_z_score	351 non-null	Float64
34	department	351 non-null	category
35	age	351 non-null	category
36	cgpa	351 non-null	category
37	academic_year	351 non-null	category
38	stress_level	351 non-null	category
39	anxiety_level	343 non-null	category
40	depression_level	351 non-null	category

dtypes: Float64(4), Int64(26), category(8), str(3)

memory usage: 109.3 KB

<class 'pandas.DataFrame'>

RangeIndex: 351 entries, 0 to 350

Data columns (total 17 columns):

#	Column	Non-Null Count	Dtype
0	age_parsed	351 non-null	Int64
1	age	351 non-null	category
2	gender	351 non-null	category

3	department	351	non-null	category
4	semester_parsed	346	non-null	Int64
5	academic_year	351	non-null	category
6	cgpa_parsed	342	non-null	Float64
7	cgpa	351	non-null	category
8	dass21_stress_score	351	non-null	Int64
9	stress_level	351	non-null	category
10	stress_z_score	351	non-null	Float64
11	dass21_anxiety_score	343	non-null	Int64
12	anxiety_level	343	non-null	category
13	anxiety_z_score	343	non-null	Float64
14	dass21_depression_score	351	non-null	Int64
15	depression_level	351	non-null	category
16	depression_z_score	351	non-null	Float64

dtypes: Float64(4), Int64(5), category(8)
memory usage: 33.1 KB

merge datasets and save outputs

```
dataset1_clean.to_csv(DATA / "dataset1_clean.csv", index=False)
dataset2_wide.to_csv(DATA / "dataset2_intermediate.csv", index=False)
dataset2_clean.to_csv(DATA / "dataset2_clean.csv", index=False)

dataset1_merge = dataset1_clean[OUTPUT_COLUMNS].copy()
dataset2_merge = dataset2_clean[OUTPUT_COLUMNS].copy()

dataset1_merge["source_dataset"] = "dataset1"
dataset2_merge["source_dataset"] = "dataset2"

# ensure "source_dataset" is categorical so that its consistent with the other cols
SOURCE_DATASET_CATEGORIES = ["dataset1", "dataset2"]
dataset1_merge["source_dataset"] = pd.Categorical(
    dataset1_merge["source_dataset"], categories=SOURCE_DATASET_CATEGORIES
)
dataset2_merge["source_dataset"] = pd.Categorical(
    dataset2_merge["source_dataset"], categories=SOURCE_DATASET_CATEGORIES
)

mhp_dataset = pd.concat([dataset1_merge, dataset2_merge], ignore_index=True)
mhp_dataset.info()
mhp_dataset.head()

mhp_dataset.to_csv(DATA / "mhp_dataset.csv", index=False)
# save to .parquet to maintain the cols dtype
mhp_dataset.to_parquet(DATA / "mhp_dataset.parquet")
```


<class 'pandas.DataFrame'>

RangeIndex: 2379 entries, 0 to 2378

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	age	2379 non-null	category
1	gender	2379 non-null	category
2	department	2379 non-null	category
3	academic_year	2379 non-null	category
4	cgpa	2379 non-null	category
5	stress_level	2379 non-null	category
6	stress_z_score	2379 non-null	Float64
7	anxiety_level	2371 non-null	category
8	anxiety_z_score	2371 non-null	Float64
9	depression_level	2379 non-null	category
10	depression_z_score	2379 non-null	Float64
11	source_dataset	2379 non-null	category

dtypes: Float64(3), category(9)

memory usage: 86.3 KB